1.54 Probability: Independence and Mutually Exclusive

Worked Solutions

1)

A and C

A and D

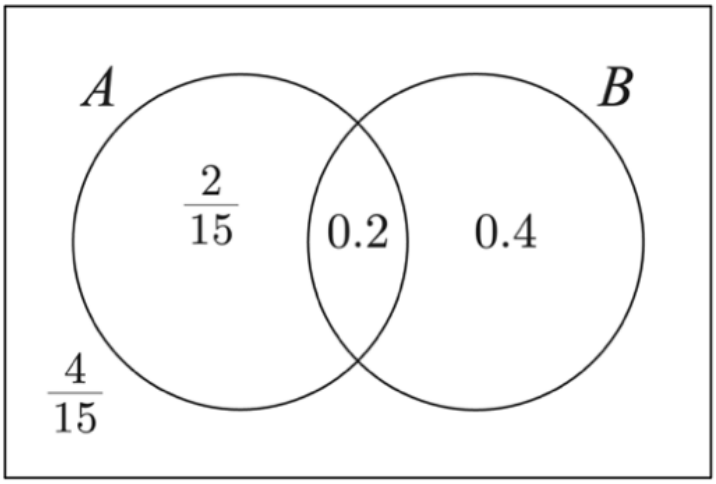
B and D

2)

A and B are independent

$$\Rightarrow \mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B) \Rightarrow 0.2 = \mathrm{P}(A) \times 0.6$$

$$\Rightarrow \mathrm{P}(A) = \frac{1}{3}$$



3)

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

$$\Rightarrow 0.95 = 0.4 + 0.55 - \mathrm{P}(A \cap B) \Rightarrow \mathrm{P}(A \cap B) = 0$$

$\therefore A$ and B are mutually exclusive events

4)

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

$$\Rightarrow 0.7 = 0.32 + 0.46 - \mathrm{P}(A \cap B)$$

$$\Rightarrow \mathrm{P}(A \cap B) = 0.08 \neq 0$$

$\therefore A$ and B are not mutually exclusive events

5)

$$\mathrm{P}(A) \times \mathrm{P}(B) = \frac{3}{7} \times \frac{2}{3} = \frac{2}{7}$$

$$\mathrm{P}(A \cap B) = \frac{2}{7}$$

$\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$ so A and B are independent events

6)

$$\mathrm{P}(A) \times \mathrm{P}(B) = \frac{3}{10} \times \frac{5}{6} = \frac{1}{4}$$

$$\mathrm{P}(A \cap B) = \frac{1}{2}$$

$\mathrm{P}(A \cap B) \neq \mathrm{P}(A) \times \mathrm{P}(B)$ so A and B are not independent events

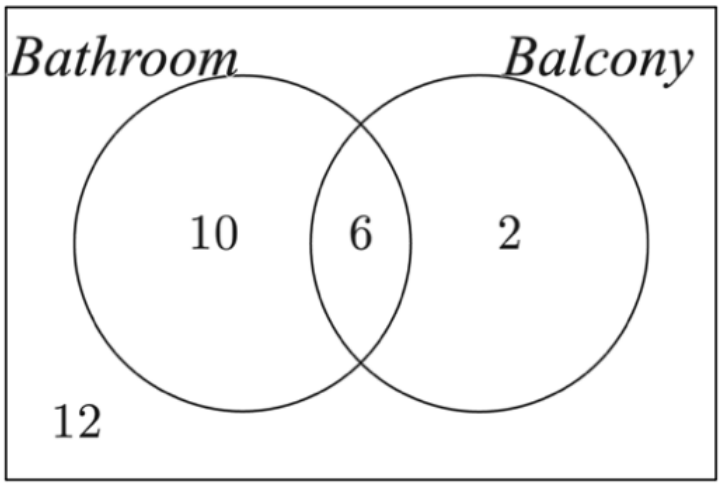
7)

$$\mathrm{P}(A) \times \mathrm{P}(B) = 0.45 \times 0.52 = 0.234$$

$$\mathrm{P}(A \cap B) = 0.23$$

$\mathrm{P}(A \cap B) \neq \mathrm{P}(A) \times \mathrm{P}(B)$ so A and B are not independent events

8)



$$\mathrm{P}(\text{Bathroom}) = \frac{16}{30}$$

$$\mathrm{P}(\text{Balcony}) = \frac{8}{30}$$

$$\mathrm{P}(\text{Bathroom}) \times \mathrm{P}(\text{Balcony}) = \frac{16}{30} \times \frac{8}{30}$$

$$= \frac{32}{225} = 0.142\ldots$$

$$\mathrm{P}(\text{Bathroom} \cap \text{Balcony}) = \frac{6}{30} = 0.2$$

$\mathrm{P}(\text{Bathroom} \cap \text{Balcony}) \neq \mathrm{P}(\text{Bathroom}) \times \mathrm{P}(\text{Balcony})$ so A and B are not independent events

9)

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

A and B are independent

$$\Rightarrow \mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$$
$$\Rightarrow 0.75 = 0.3 + \mathrm{P}(B) - 0.3 \times \mathrm{P}(B)$$
$$\Rightarrow 0.45 = 0.7\mathrm{P}(B) \Rightarrow \mathrm{P}(B) = \frac{9}{14}$$

10)

A and B are independent

$$\Rightarrow \mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$$
$$\Rightarrow 0.85 = 0.95 \times \mathrm{P}(B) \Rightarrow \mathrm{P}(B) = \frac{17}{19}$$
$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$
$$\Rightarrow \mathrm{P}(A \cup B) = 0.95 + \frac{17}{19} - 0.85 = \frac{189}{190}$$

11)

$$1 = k + k + \frac{3}{20} + \frac{3}{10} + \frac{19}{70}$$
$$\Rightarrow 1 = 2k + \frac{101}{140} \Rightarrow k = \frac{39}{280}$$
$$\mathrm{P}(A) = \frac{39}{280} + \frac{3}{20} + \frac{3}{10} = \frac{33}{56}$$
$$\mathrm{P}(B) = \frac{3}{10} + \frac{19}{70} = \frac{4}{7}$$
$$\mathrm{P}(A) \times \mathrm{P}(B) = \frac{33}{56} \times \frac{4}{7} = \frac{33}{98} = 0.336\ldots$$
$$\mathrm{P}(A \cap B) = \frac{3}{10} = 0.3$$

$\mathrm{P}(A \cap B) \neq \mathrm{P}(A) \times \mathrm{P}(B)$ so A and B are not independent events

12)

$$1 = \frac{31}{280} + \frac{1}{20} + k + k + p \Rightarrow p = \frac{47}{56} - 2k$$
$$\mathrm{P}(A \cap C) = \mathrm{P}(A) \times \mathrm{P}(C)$$
$$\Rightarrow k = \left(\frac{1}{20} + k\right)(k + p)$$
$$\Rightarrow k = \left(\frac{1}{20} + k\right)\left(k + \frac{47}{56} - 2k\right)$$
$$\Rightarrow k = \left(\frac{1}{20} + k\right)\left(\frac{47}{56} - k\right)$$
$$\Rightarrow k = \frac{47}{1120} - \frac{1}{20}k + \frac{47}{56}k - k^2$$

$$\Rightarrow k^2 + \frac{59}{280}k - \frac{47}{1120} = 0$$
$$\Rightarrow k = \frac{1}{8} \text{ as } k \neq -\frac{47}{140}$$
$$\therefore p = \frac{47}{56} - 2 \times \frac{1}{8} = \frac{33}{56}$$

13)

B and C are mutually exclusive so $z = 0$

$$1 = 0.24 + x + 0.2 + y + 0.2 + 0.06$$
$$\Rightarrow x + y = 0.3 \Rightarrow y = 0.3 - x$$

A and B are independent

$$\Rightarrow \mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$$
$$\Rightarrow 0.2 = (x + 0.4)(y + 0.2)$$
$$\Rightarrow 0.2 = (x + 0.4)(0.3 - x + 0.2)$$
$$\Rightarrow 0.2 = (x + 0.4)(0.5 - x)$$
$$\Rightarrow 0.2 = -x^2 + 0.5x - 0.4x + 0.2$$
$$\Rightarrow x^2 - 0.1x = 0 \Rightarrow x(x - 0.1) = 0$$
$$\Rightarrow x = 0.1 \text{ or } x = 0$$

If $x = 0.1$, then $y = 0.3 - 0.1 = 0.2$

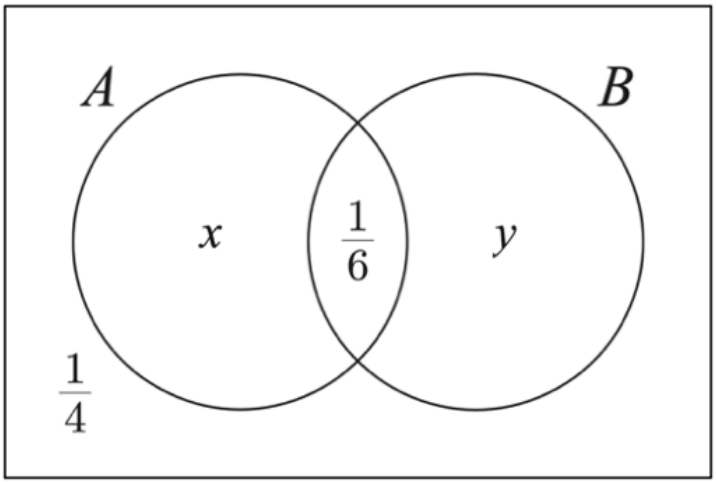
If $x = 0$, then $y = 0.3 - 0 = 0.3$

$$\therefore x = 0.1,\ y = 0.2,\ z = 0$$

or $x = 0$, $y = 0.3$, $z = 0$

14)

$$\mathrm{P}((A \cup B)') = 1 - \mathrm{P}(A \cup B) = 1 - \frac{3}{4} = \frac{1}{4}$$



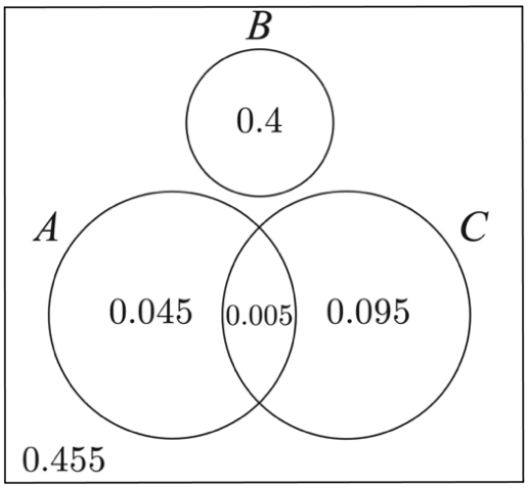
$$1 = \frac{1}{4} + x + \frac{1}{6} + y \Rightarrow y = \frac{7}{12} - x$$

A and B are independent

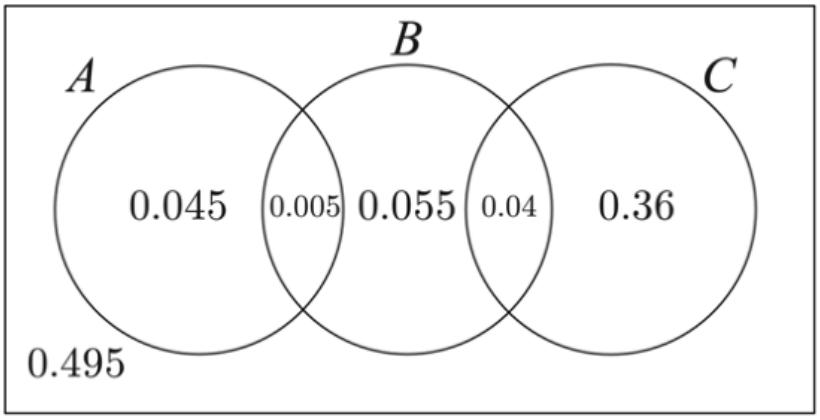
$$\Rightarrow \mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$$
$$\Rightarrow \frac{1}{6} = \left(x + \frac{1}{6}\right)\left(\frac{1}{6} + y\right)$$
$$\Rightarrow \frac{1}{6} = \left(x + \frac{1}{6}\right)\left(\frac{1}{6} + \frac{7}{12} - x\right)$$
$$\Rightarrow \frac{1}{6} = \left(x + \frac{1}{6}\right)\left(\frac{3}{4} - x\right)$$
$$\Rightarrow \frac{1}{6} = -x^2 + \frac{3}{4}x - \frac{1}{6}x + \frac{1}{8}$$
$$\Rightarrow x^2 - \frac{7}{12}x + \frac{1}{24} = 0$$
$$\Rightarrow x = \frac{1}{12} \text{ or } x = \frac{1}{2}$$

If $x = \frac{1}{12}$, then $y = \frac{7}{12} - \frac{1}{12} = \frac{1}{2}$
If $x = \frac{1}{2}$, then $y = \frac{7}{12} - \frac{1}{2} = \frac{1}{12}$
$\therefore \mathrm{P}(A) = \frac{1}{12} + \frac{1}{6} = \frac{1}{4}$ and $\mathrm{P}(B) = \frac{1}{6} + \frac{1}{2} = \frac{2}{3}$
Or $\mathrm{P}(A) = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$ and $\mathrm{P}(B) = \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$

15)



16)



17)

$$\begin{aligned} \mathrm{P}(A) &= 0.0522 + 0.3078 = 0.36 \\ \mathrm{P}(B) &= 0.0522 + 0.0928 = 0.145 \\ \mathrm{P}(A \cap B) &= 0.0522 \\ \mathrm{P}(A) \times \mathrm{P}(B) &= 0.36 \times 0.145 = 0.0522 = \mathrm{P}(A \cap B) \end{aligned}$$

$\therefore A$ and B are independent events.

18 a)

$\mathrm{P}(B) = \frac{24}{75}$

18 b)

$$\begin{aligned} \mathrm{P}(A \cup B) &= \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B) \\ &= \frac{52}{75} + \frac{24}{75} - \frac{14}{75} = \frac{62}{75} \end{aligned}$$

18 c)

$\mathrm{P}(A \cap B') = \frac{20+18}{75} = \frac{38}{75}$

18 d)

$$\begin{aligned} \mathrm{P}(A) = \frac{52}{75},\ \mathrm{P}(B) = \frac{24}{75},\ \mathrm{P}(A \cap B) = \frac{14}{75} \\ \mathrm{P}(A) \times \mathrm{P}(B) = \frac{52}{75} \times \frac{24}{75} = \frac{416}{1875} \neq \frac{14}{75} = \mathrm{P}(A \cap B) \end{aligned}$$

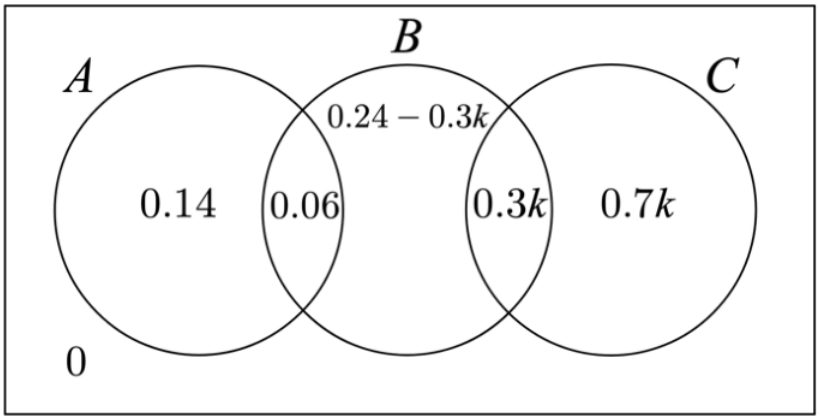
so A and B are not independent events

19)

$$\begin{aligned} \mathrm{P}(A' \cap B) &= 0.12 + 0.04 = 0.16 \\ \mathrm{P}(C) &= 0.04 + 0.1 + 0.09 + 0.02 = 0.25 \\ \mathrm{P}((A' \cap B) \cap C) &= 0.04 \\ \mathrm{P}(A' \cap B) \times \mathrm{P}(C) &= 0.16 \times 0.25 = 0.04 \\ &= \mathrm{P}((A' \cap B) \cap C) \end{aligned}$$

so $A' \cap B$ and C are independent events

20)



$$0.14 + 0.06 + (0.24 - 0.3k) + 0.3k + 0.7k = 1$$

$$\Rightarrow 0.44 + 0.7k = 1 \Rightarrow k = 0.8$$